

Project Proposal

Project Title:

Secure CNIC & Face Verified Voting System (Blockchain-Inspired)

Introduction:

This project aims to develop a secure and transparent voting system that ensures each vote is authentic and unique by using user identity verification through CNIC and real-time face capture. Traditional voting systems are vulnerable to fraud multiple voting and data tampering. Our proposed system solves these issues by using blockchain-inspired data integrity and biometric verification ensuring that only verified voters can cast their votes.

Problem Statement:

In many voting systems fake or multiple votes can be cast using false identities and there is often no reliable verification process to ensure transparency. Additionally manual counting is time-consuming and error-prone. Therefore there is a need for a secure digital and verifiable system that guarantees one-person-one-vote and prevents manipulation.

Proposed Solution:

The proposed system provides a GUI-based voting application developed in Python that:

1. Allows each voter to enter their CNIC number .
2. Requires the voter to upload a picture of their CNIC .
3. Uses the system webcam to capture the voter's live photo for identity verification.
4. Once verified the user can select a candidate and cast their vote.
5. Votes are securely stored in a tamper-proof structure simulating blockchain principles (no vote can be changed or deleted).
6. The system displays the total results and the winner after voting ends.

System Features:

CNIC Verification: Each CNIC can only vote once.

Image Upload: User must upload CNIC photo.

Face Capture: Uses webcam to capture live face image.

Vote Recording: Each verified vote is stored securely.

Result Display: Real-time results with total votes per candidate.

GUI Interface: User-friendly and interactive design (Tkinter).

Tools & Technologies:

Component	Technology
Programming Language	Python
GUI Framework	Tkinter
Image Processing	OpenCV
Image Handling	Pillow (PIL)
Platform	Windows / Linux

Expected Outcome:

A fully functional voting system with user verification and result calculation.

Demonstration of how blockchain concepts like immutability and transparency can be applied in a

simple digital voting scenario. A secure efficient and user-friendly application suitable for educational or small-scale election purposes.

Future Enhancements:

Integration with real blockchain (Ethereum / Hyperledger) for permanent record storage.

Face recognition matching using AI instead of simple capture. Cloud-based deployment for multiple users. Admin dashboard for live monitoring of votes.

Conclusion:

This project presents a step towards secure and verified digital voting using modern technologies. By combining blockchain concepts CNIC verification and biometric verification it offers transparent and reliable solution to modern voting challenges.

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